1 HR Master Transfer

This interface for transferring HR data to HYDRA has been implemented using a new technology for universal interfaces. Using this technology, a separate command (so-called "dialogs") is listed in the interface file for each line to import the required data. The interface program processes these dialogs sequentially and writes a log file of the results.

Each line of the interface file contains a dialog with all of the relevant data. The most important components are the **dialog ID** and the **parameters** belonging to the dialog.

Example of a dialog (one line from an interface file):

DLG=PNR.DELETE|PNR.FIR=012|PNR.PNR=002449|...

Explanation

PNR.DELETE is the dialog identification. Vertical lines ("|", ASCII 124) are used to separate the different parameters (PNR.FIR, PNR.PNR, ...). Select a field width for the different parameters that is wide enough for the presentation. However a fixed file structure can also be realized by adding leading zeros (for numbers) and trailing spaces. The sequence of the parameters is irrelevant. Some parameters are mandatory and must always be included in a dialog. Other parameters are optional and are specified if required or left out. Fields that are not included in the interface are assigned the default value when a person is created. These fields are not overwritten when you change the data of a person. You therefore manage these fields in HYDRA.

1.1 Character set

As of MESWeaver 2.1, HYDRA is based on Unicode. Interface files are therefore expected in format UTF-8 without BOM (Byte Order Mark).

1.2 Formatting of different data types

The following formats are supported:

Data type	Format	Examples
N<x>	Digits, maximum of <x> places	|PNR.PNR=2449| or |PNR.PNR=002449|
N<x>.<y>	Decimal number with a maximum of <x> predecimal places and <y> decimal places. A dot is the decimal separator.	|PNR.PSTDSATZ=30.5| or |PNR.PSTDSATZ=00030.5|

<table>
<tr><th>Data type</th><th>Format</th><th>Examples</th></tr>
<tr><td>C<x>
Texts (character)</td><td>Character string with the maximum length <x>; the maximum length does not need to be filled up with blank characters.</td><td>`|PNR.NAME=Huber|` or
`|PNR.NAME=Huber      |`</td></tr>
<tr><td>Date</td><td>MM/DD/YYYY (American format: month, day, year. Slashes are used as separator.)</td><td>`|STMP.DAT=12/31/2002|`</td></tr>
<tr><td>Times or durations</td><td>Seconds since midnight or
HH:MM or
HH:MM:SS or
HH,DDD or
HH.DDD

H hours (as many places as required)
M Minutes (in groups of 60)
S Seconds
D Industrial or decimal minutes (in groups of 100)</td><td>`|STMP.ZEI=52200|` or
`|STMP.ZEI=14:30|` or
`|STMP.ZEI=14:30:00|` or
`|STMP.ZEI=014,5|` or
`|STMP.ZEI=14.500|`</td></tr>
</table>

1.3 Available dialogs

The following dialogs are available to transfer the persons:

- PNR.INSERT to create a person
- PNR.UPDATE to change a person
- PNR.DELETE to delete a person
- PNR.MODIFY This dialog checks on the basis of the personnel number whether or not the person already exists in HYDRA. If the person is available, the person is modified, if not, the person is created.

Note:

If a person is created using PNR.INSERT or PNR.MODIFY, the system automatically assigns basic authorizations for the clockings of the time and attendance (PZE) and for the access of the access control (ZKS). In PZE, the person is authorized for the terminal group 99, and in ZKS, the access profile 999 is assigned automatically.

1.4 Parameters for the transfer of HR master data

The identifiers in column "Mandatory" specify the dialogs with mandatory fields:

I For INSERT and MODIFY

U For UPDATE and MODIFY

D FOR DELETE

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
General data				
PNR.PNR	N8	I/U/D	Personnel number	The personnel number is the key field to access the person's data
PNR.PNAME	C 40		Name	
PNR.PVORNAME	C 20		First name	
PNR.PVORNAME:2	C 20		Middle name	
PNR.KUERZEL	C10		Initials	Initials of the person (*1)
PNR.FIR	C 4	I	Company	Company of the person
PNR.BER	C 8	I	Area	
PNR.KST	C10	I	Cost center	The person's regular cost center
PNR.EINTRITT	Date	I	Date of joining	
PNR.AUSTRITT	Date		Date of leaving	
PNR.SVERTRETER1	Integer		Replacement 1	Personnel number of replacement 1 (*1)
PNR.SVERTRETER2	Integer		Replacement 2	Personnel number of replacement 2 (*1)
PNR.KNR	C10		Staff badge	Badge number
PNR.ANREDE	C 20		Salutation	Salutation of the person (*1)
PNR.PRODEMPLOYEE	C1		Production employees	Identifies if it is a production employee or not (available as of SP13/25.10.2018)

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PNR.VAB	C 15		Responsibility area	ID controlling which user is authorized to access the data of which persons. If the responsibility area remains empty, every user can access the person's data.
PNR.TAETIGKEIT	C 20		Activity	
PNR.TITEL	C 20		Title	Academic titles (form of address)
PNR.NATION	C 3		Nationality	e.g. "D" or "F" or "CH" or "GB" or "US" or "CZ",etc.
PNR.GEBDAT	Date		Date of birth	
PNR.GEBORT	C 30		Place of birth	Place of birth of a person (*1)
PNR.SCHULE1	C 50		School-leaving qualification	School-leaving qualification of the person (*1)
PNR.SCHULE2	C 50		Secondary school-leaving qualification	Secondary school-leaving qualification of the person (*1)
PNR.STRASSE	C50 (*1)		Street	Street and street number of place of residence
PNR.PLZ	N5		Zip code	ZIP code of the place of residence
PNR.ORT:WOHN	C 20		Domicile	
PNR.TEL:FIR	C 20		Company phone	
PNR.TEL:PRIVAT	C 20		Private phone	
PNR.MOBILTEL:FIR	C 20		Company mobile	
PNR.MOBILTEL:PRIVAT	C 20		Private mobile	
PNR.EMAIL:FIR	C 50		Company e-mail	
PNR.EMAIL:PRIVAT	C 50		Private e-mail	
PNR.GESCHLECHT	C1		Gender	M: male W: female
PNR.FAMSTAND	C1		Family status	L: Single V: Married W: Widowed G: Divorced
PNR.PNR:VGS	N8		Supervisor	Personnel number of supervisor
PNR.INFOTXT:n	C 40		Text field n (n from 1 to 10)	Free text field with a maximum of 40 characters. In particular with existing or planned interfaces to SAP, pay attention to the notes on the info fields below.
PNR.INFOTXT:n	C 20		Text field n (n from 11 to 15)	Free text field with a maximum of 20 characters. In particular with existing or planned interfaces to SAP, pay attention to the notes on the info fields below.
PNR.INFOTXT:n	C10		Text field n (n from 16 to 20)	Free text field with a maximum of 10 characters. In particular with existing or planned interfaces to SAP, pay attention to the notes on the info fields below.

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PNR.INFOWERT:n	N8		Number field n (n from 1 to 5)	Free number field with a maximum of 8 characters. In particular with existing or planned interfaces to SAP, pay attention to the notes on the info fields below.
PNR.INFODAT:n	Date		Date n (n from 1 to 5)	Free date field. In particular with existing or planned interfaces to SAP, pay attention to the notes on the info fields below.
Data for the Personnel Time Management PZW				
PNR.ABT	C 8		Department	
PNR.PKREIS	C 8		Employee subgroup	
PNR.GLZJMOD	N4		Working time model	Number of the model
PNR.SCHZARTJMOD	N4		Shift rhythm model	Number of the model
PNR.ENTLJMOD	N4		Payment model	Number of the model
PNR.ENTLTMOD:MEHRARB	N4		Overtime type	Number of the payment day type
PNR.BESCHVERH	C1		Type of contract	G: Salaried A: Non-salaried
PNR.AVGAZ	Duration		Average working time	If configured accordingly, this time is posted for absences.
PNR.TZGRAD	N3.3		Part-time rate	Part-time rate in percent with a maximum of three decimal places
PNR.ETGAWDAT	Date		First allocation	Date when this person is first evaluated by the PZE workday evaluation. You can fill in this field if the date is not the date of joining.
PNR.ZNWL	N3		Time sheet	Number of time sheet for display in SMA or WEB
PNR.DGBERECHT	C1		Business trip authorization	J/N
PNR.SPERR:PZE	C1		Blocking indicator PZE	S: Person is blocked for PZE. Empty: Person is not blocked.
PNR.OPT:NSTMP	C1		Person does not clock	J/N
PNR.OPT: AVGAZVERB	C1		Allocate average working time	J/N
PNR.URLANSPR: NORM	N3.1		Annual leave entitlement	Annual entitlement to vacation.
PNR.URLANSPR:SONDER	N3.1		Special leave entitlement	Annual entitlement to special leave.
PNR.URLANSPR: ZUSATZ	N3.1		Additional leave entitlement	Annual entitlement to additional leave.
Data for Shop Floor Data Collection BDE				
PNR.PGRP	N3		Employee group	
PNR.BDEJMOD [PNR.SJMOD:BDE]	N3		Year model	Number of the BDE shift year model (The ID PNR.SJMOD:BDE is out-dated and only used if PNR.BDEJMOD is not set.)

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PNR.SMNR	C 8		Workplace	Regular workplace. Leading zeros are required for numeric machine numbers!
PNR.MEHRMNR	C1		Multi machine operation	Logon to multiple machines allowed J/N.
PNR.OPT:AGWAUTO	C1		Automatic OP change	J/N
PNR.PLAUS: PNRANAG	C1		BDE check whether or not the person has to be logged on to the OP	J/N
PNR.PLAUS: SMENGE	N1		Activate BDE target quantity check	0: The operation can be logged off at any time 1: A check is performed indicating whether or not the current actual quantity is within the "underproduction" and "overproduction" limits. If this is not the case the logon is rejected. 2: as in 1, in addition the person has to be logged on to the OP!
PNR.UPG	N3		Minimum target quantity	For the target quantity check, in percent of the planned quantity.
PNR.OPG	N3		Maximum target quantity	For the target quantity check, in percent of the planned quantity.
PNR.OPT:PABSKE	C1		Automatic logoff of personnel when shift ends	J/N. Default=J. This option is used if the automatic shift change function is in use and if the person is logged on to an operation where the option *Use workplace settings* is defined for the relevant order type and if the person is logged on to a workplace where the *Use personal settings* option is set.
PNR.ASTUFE:1	N1		Authorization for orders	0..9: Authorization level specifying whether or not the person may log on orders (comparison with the authorization level of the operation). see below.
PNR.ASTUFE:2	C1		Log OP on	J: The above authorization level refers to the logon of operations.
PNR.ASTUFE:3	C1		Log OP off	J: The above authorization level refers to the logoff of operations.
PNR.ASTUFE:4	C1		Log off all staff	J: The person may execute the terminal function "log all people off".
PNR.MSTUFE:1	N1		Authorization for machine status changes	0..9: Authorization level specifying whether or not the person is allowed to change statuses (comparison of the authorization level from the (machine) status assignment).

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PNR.MSTUFE:2	C1		Change only if person is logged on	The person may only change statuses on the terminal if the person is currently logged on to the workplace.
PNR.MSTUFE:4	C1		Change of production lock (from HYDRA-MDE 7.2 on)	J: Authorization available N: No authorization By default = N.
PNR.SSTUFE:1	N1		Authorization to change target cycle and target partitioning on the terminal	0: No authorization 1: Authorization available
PNR.SSTUFE:2	C1		Authorization to change the target quantity on the terminal	J: Authorization available N: No authorization
PNR.SSTUFE:2	C1		Authorization to change the target quantity on the terminal	J: Authorization available N: No authorization
PNR.RSTUFE	N1		Status change of resources	0..9: Authorization level specifying whether or not the person is allowed to change resource statuses (comparison with the authorization level from the resource status assignment). Only relevant if HYDRA WRM or HYDRA DNC is in use
PNR.DLSTUFE	N1		DNC download authorization	0 = No authorization 9 = Authorization available Only relevant if HYDRA DNC is in use
PNR.ULSTUFE	N1		DNC upload authorization	0 = No authorization 9 = Authorization available Only relevant if HYDRA DNC is in use
PNR.SPERR:BDE	C1		Blocking ID BDE	S: Person is blocked for BDE (shop floor data collection). Empty: Person is not blocked. Please note: The field is NOT visible on the console.
Data for the Incentive Wage LLE				
PNR.PRKZ	C1		Premium indicator	'E' = individual piecework, 'G' = group piecework 'Z' or empty = time wage Default: empty.
PNR.PRGRP	C 3		Premium group •	
PNR.ANTFAKTLBON	N3		Premium factor	

Dialog: PNR.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
PNR.BPOS	C 6		Operator position/function	
PNR.LPKZ	C10		Wage/premium indicator	
PNR.LART	C 4		Wage type	
PNR.LGRP	C 4		Wage group	
PNR.OPT:PZEADEABGL	C1		BDE/PZE comparison	J/N

(*1): These fields and the respective parameters are only available, if the extension persfieldsPZW83 is activated (available since 01-MAR-2019).

> **Note:**
>
> When the persons are initially created (first interface run), do not pass the parameters PNR.PNR:VGS(supervisor); PNR.SVERTRETER1 (replacement 1); PNR.SVERTRETER2 (replacement 2) because this can lead to errors. An error occurs if a person is assigned to a supervisor that comes later in the interface file and is therefore not yet created. Recommendation: perform two inface runs one after the other. The first run is used to create the persons without supervisor. It guarantees that all persons are available. The second run is used to assign the supervisor. This run includes the parameters PNR.PNR:VGS(supervisor); PNR.SVERTRETER1 (replacement 1); PNR.SVERTRETER2 (replacement 2).

1.5 Notes on some parameters

1.5.1 Personnel number PNR.PNR

Unique number of the person. The personnel number has a maximum length of eight characters.

1.5.2 Badge PNR.KNR

Number on the staff badge that is assigned to the person. HYDRA can process badge numbers that are up to 10 characters long, however the number of characters considerably depends on the ID badge in use. If barcode badges are used for identification, numbers that are four characters long are recommended. The number of characters for the badge number may be defined in HYDRA, i.e. the interface program only transfers the configured number of characters. If the badge number is shorter than defined in the HYDRA setup, it is filled with leading zeros. If it is too long an error message appears and the dialog is rejected.

> **Note:**
>
> The badge number of a person can be empty. If a number is assigned, it must be unique in HYDRA like the personnel number. If a badge number is passed for a person that had been assigned to a person that has already left the company, the badge number of the person who has left is deleted and assigned to the new person.

1.5.3 Authorization levels PNR.ASTUFE and PNR.MSTUFE

Authorization logic that may be defined individually for ADE and MDE activities on the shop floor terminals. 0 = no authorization, .1 = low authorization, 9 = highest authorization.

1.5.4 Info fields

If your system has interfaces to SAP or will have SAP interfaces at a later stage, then please mind that part of the info fields is assigned SAP-specific additional data when the MINI HR master data is downloaded from SAP-HCM. Some fields are also required to control the person-related upload of data to the correct SAP system.

Free info text 9

When downloading the MINI HR master data from SAP, this field is assigned the "Customer Field 2" from SAP. By default, this field is not used by HYDRA.

Free info text 14

When downloading the MINI HR master data from SAP, this field is assigned the "Customer Field 1" from SAP. By default, this field is not used by HYDRA.

Free info text 16

When downloading the MINI HR master data from SAP, this field is assigned the "Source System" from SAP.

When uploading person-related data via interfaces to SAP, the Source System heads for the target SAP system the data of the person is transferred to. This applies for the following interfaces:

- Uploading time events via HR-PDC
- Uploading data of the Personnel Time Management to SAP-HCM, e.g. wage type postings or absences
- Uploading data of the Incentive Wage to SAP-HCM
- Other customer-specific interfaces with person-related data to SAP in the HR context (not PP-PDC)

If these interfaces are used, the customer must ensure that no other field contents are assigned to the info text 16. The info text 16 can remain empty, then the source system is identified using other methods; see documentation of the relevant interface.

Free info text 17

When downloading the MINI HR master data from SAP, this field is assigned the "Country Grouping" from SAP. This field can be used in HYDRA with the Time and Attendance PZE as subsystem of SAP-HCM (HR-PDC) when data is collected in combination with plausibility checks.

Free info text 18

When downloading the MINI HR master data from SAP, this field is assigned the field "ES_GRPG_WORK_SCHED" from SAP. This field can be used in HYDRA with the Time and Attendance PZE as subsystem of SAP-HCM (HR-PDC) when data is collected in combination with plausibility checks.

Free info text 19

When downloading the MINI HR master data from SAP, this field is assigned a content that is made up of the fields PS_GRPG_ATT_ABS_TYPE and ATT_ABS_REASON_GRPG. This field can be used in HYDRA with the Time and Attendance PZE as subsystem of SAP-HCM (HR-PDC) when data is collected in combination with plausibility checks.

Free info text 20

When downloading the MINI HR master data from SAP, this field is assigned the field "EXT_WAGETYPE_GRPG" from SAP. This field can be used in HYDRA with the Time and Attendance PZE as subsystem of SAP-HCM (HR-PDC) when data is collected in combination with plausibility checks.

1.6 HR master with version control

1.6.1 Overview

Optionally, you can manage the HYDRA HR master data in versions. You can manage different versions of a person that are/were valid at different times. To this end, the additional parameter **PNR.DATB** is used as "validity start date". It includes the validity start date in the date format.

The unique key for a HR master version consists of the personnel number PNR.PNR and the validity start date PNR.DATB. This applies for the update and deletion of HR master versions.

HYDRA automatically manages the validity end date of a version. An HR master version applies until the next version becomes effective. The validity date is not restricted if there is no subsequent version.

In case the validity start date is not specified, the interface always refers to the HR master version that is in effect at the time when the interface process is running.

1.6.2 Please note for PNR.MODIFY

If the validity start date PNR.DATB is not specified for the PRN.MODIFY dialog and there is currently no applicable HR master version, a currently applicable HR master version will be added by using the parameters transferred. Moreover, all existing HR master versions that are applicable from this day on are updated by the parameters transferred.

If the validity start date PNR.DATB is specified for the PNR.MODIFY dialog and this HR master version does not yet exist in HYDRA, this version will be added and an already existing version that is applicable on the validity start date will be used as basis (copied) and updated by the transferred parameters.

1.7 Sample data record for a person

All fields affecting a person are written in one line in the interface. In the below example, however, data is written in two lines to make it readable:

```
DLG=PNR.MODIFY|PNR.PNR=153443|PNR.KNR=001324|PNR.FIR=BSP|PNR.KST=B12_54|
PNR.BER=Halle 17|PNR.EINTRITT=01/15/2010|PNR.PNAME=Maier|PNR.PVORNAME=Hans|
```

1.8 Import of data in HYDRA

1.8.1 Manual start of the interface program

Existing personnel data can be processed using the program hymw.

Windows: `hymw -u 199 -b filename.bap > filename.log`

or

Linux: `hymw.out -u 199 -b filename.bap > filename.log`

Note: You must call the program in the HYDRA directory.

Parameter

-u <usernummer>

Here, a unique numeric value must be selected as the <UserNumber>. If interfaces run at the same time with the same UserNumber, they interfere with each other and supply false results. Recommended are numbers ranging between 1 to 200.

-b <inputFile>

Name of file with HR master data.

> <logfile>

If you redirect the output, the GUI output is redirected to a log file. The processing results can then be analyzed later on.

Example

Sample file myFile.bap

```
DLG=PNR.MODIFY|PNR.PNR=11111111|PNR.KNR=1111|PNR.FIR=BSP|PNR.KST=costc_001|PNR.BER=Area57|PNR.EINTRITT=01/15/2019|PNR.PNAME=Miller|PNR.PVORNAME=Heinz|
DLG=PNR.MODIFY|PNR.PNR=12345678|PNR.KNR=001324|PNR.FIR=BSP|PNR.KST=B12_54|PNR.BER=Area17|PNR.EINTRITT=12/01/2017|PNR.PNAME=Maier|PNR.PVORNAME=Hans|
```

Request via command line

```
hydadm:2:E:\hydra2>hymw.exe -u999 -bmyFile.bap > myFile.log

hydadm:2:E:\hydra2>
```

Log file myFile.log

```
EXEC:DLG=PNR.MODIFY|PNR.PNR=111111|PNR.KNR=1111|PNR.FIR=BSP|PNR.KST=costc_001|PNR.BER=Area57|PNR.EINTRITT=01/15/2019|PNR.PNAME=Miller|PNR.PVORNAME=Heinz|
RES: RET=0|KT=|LT=|RET=0|KT=|LT=|
EXEC:DLG=PNR.MODIFY|PNR.PNR=153443|PNR.KNR=001324|PNR.FIR=BSP|PNR.KST=B12_54|PNR.BER=Area17|PNR.EINTRITT=12/01/2017|PNR.PNAME=Maier|PNR.PVORNAME=Hans|
RES: RET=1703|KT=Invalid badge no.|LT=Invalid badge number 001324            |
```

The first data record has been processed by the system without error ("RET=0|"). The second data record produced an error (RET=1703|). The following texts shortly describe the error.

To change the language of the error texts, set the environment variable HYLANG in the command line to "en" or "de" before you execute the command:

```
hydadm:2:E:\hydra2>set HYLANG=en
```

or

```
hydadm:2:E:\hydra2>set HYLANG=de
```

1.8.2 Definition for transferring data from SAP

The below structure has been designed to transfer HR master data from SAP in the HYDRA BAPI format using a customer-specific function module.

Message type:	ZHYDRA_PERSONS
IDoc type:	ZHYDRA_PERSONS01
Segments:	Z2BAPI000

NOTE

To generate segment names in HYDRA inbound processing as described above, the segments must have been created in SAP according to the pattern Z1<segment name>. The SAP outbound processing then generates versions using the segment names Z2<segment name><version>.

Example: **Z1**BAPI becomes **Z2**BAPI**000**

Field name	Type		Description	Example
Transaction	CHAR	20	Transaction ID (dialog ID in HYDRA)	PNR.MODIFY
Description	CHAR	40	Plain text designation as comment	Download HR master
Data	CHAR	940	Dialog data string for HYDRA	DLG=PNR.MODIFY|PNR.PNR=12345678| PNR.KNR=00000001| Details see section 2.3.1

1.8.3 Transfer from other systems (file-based via MLE)

The communication between HYDRA and higher level systems generally depends on the technical configuration and capabilities of the corresponding system. HYDRA provides the RFC technology or the classic transfer option using ASCII files (“Text Files” = .txt files) for the communication (to and from HYDRA).

"IDOCs" combine data to logical units within one file. These IDOCs act like a “bracket” and combine logically similar data structures to transfer several of these “clusters” within one file. Although each IDOC corresponds to a defined data type/structure, the format does not depend on the content or the content type.

For details on the configuration of the interface, refer to the document "MES Weaver" (SIS-MWV_30.PDF).

Data is transferred using the following basic structure:

Field name	Type	Length	Designation (name)	Data field and meaning
SEGNAM*	Char	30	Segment	To this field, the writing system assigns the respective segment name. This segment name distinctly defines the structure of the data record (field SDATA). Example: Z2BAPI000
MANDT*	Char	3	Instance	Reserved; fixed: '000'
DOCNUM*	Char	16	IDoc number	Consecutive number for IDOCs Reserved: fixed '0000000000000000'
SEGNUM*	Char	6	Segment number	Reserved: fixed '000000'
PSEGNUM	Char	6	Number of the parent segment	Reserved; fixed: '000000'
HLEVEL	Char	2	Hierarchy level	Reserved; fixed: '00'
SDATA	Char	1000	Payload	This field contains the actual data content/payload. The SEGNAM field specifies the structure of this field.

* = Key field

The below structure has been designed to transfer HR master data from other systems using the HYDRA file interface "File Port":

Message type / file name:	ZHYDRA_PERSONS
Message function / file extension:	.dat
Segments:	Z2BAPI000

The segment Z2BAPI000 has the following structure:

Field name	Type		Description	Example
Transaction	CHAR	20	Transaction ID (dialog ID in HYDRA)	PNR.MODIFY
Description	CHAR	40	Plain text designation as comment	Download HR master
Data	CHAR	940	Dialog data string for HYDRA	DLG=PNR.MODIFY|PNR.PNR=12345678| PNR.KNR=00000001| Details see section 2.3.1